

Predicting Monetary Policy Stance from FOMC Minutes

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ABSTRACT

This project builds an AI/NLP pipeline that converts public Federal Reserve FOMC meeting minutes into a document-level hawkish-dovish policy stance index. It compares a transparent dictionary baseline with a FinBERT classifier trained from dictionary-generated weak labels. Both classifiers are evaluated against an independent 30-sentence human audit. The resulting tone index is also compared descriptively with U.S. Treasury yields across monetary-policy regimes. The market analysis is exploratory, not causal, and not a minutes-release event study.

1. Introduction

PROBLEM STATEMENT

Central-bank communication moves financial markets because minutes and statements reveal policymakers' assessment of inflation, employment, financial conditions, and future policy risks. The practical challenge is that FOMC minutes are long, qualitative documents. A systematic NLP index can make policy tone comparable across meetings and across decades.

RESEARCH GOALS

- Estimate whether each FOMC minute is relatively hawkish or dovish.
- Aggregate sentence classifications into a comparable document-level tone index.
- Evaluate the dictionary and FinBERT classifiers against independent human labels.
- Examine whether the tone index comoves descriptively with Treasury yields.

EXPECTED RELATIONSHIP

Hawkish language should generally align with higher expected policy rates and higher short-to-intermediate yields, while dovish language should align with easing expectations. The project contributes a reproducible pipeline using public data, interpretable sentence-level labels, and an exploratory market association analysis.

2. Background

RELATED WORK

- Loughran and McDonald (2011) show why domain-specific dictionaries improve financial tone measurement over general sentiment dictionaries.
- Lucca and Trebbi (2009) and Hansen and McMahon (2016) show that central-bank text contains measurable signals relevant to asset prices and expectations.
- BERT (Devlin et al., 2019) and FinBERT (Araci, 2019) motivate the transformer-model comparison.

AI/ML CONCEPTS USED

- Dictionary classification assigns each sentence a hawkish, dovish, or neutral label from monetary-policy term counts.
- Document-level aggregation converts sentence labels into a continuous tone index.
- FinBERT is fine-tuned on dictionary-generated labels, a form of weak supervision.
- A blind 30-sentence human audit independently evaluates both classifiers.
- A separate Treasury-yield analysis examines exploratory market association.

3. Methodology

TOOLS AND FRAMEWORKS

- Data and collection: Python, pandas, NumPy, BeautifulSoup, and requests/curl.
- Modeling and evaluation: scikit-learn, SciPy, PyTorch, and Hugging Face Transformers.
- Reporting and visualization: Matplotlib.

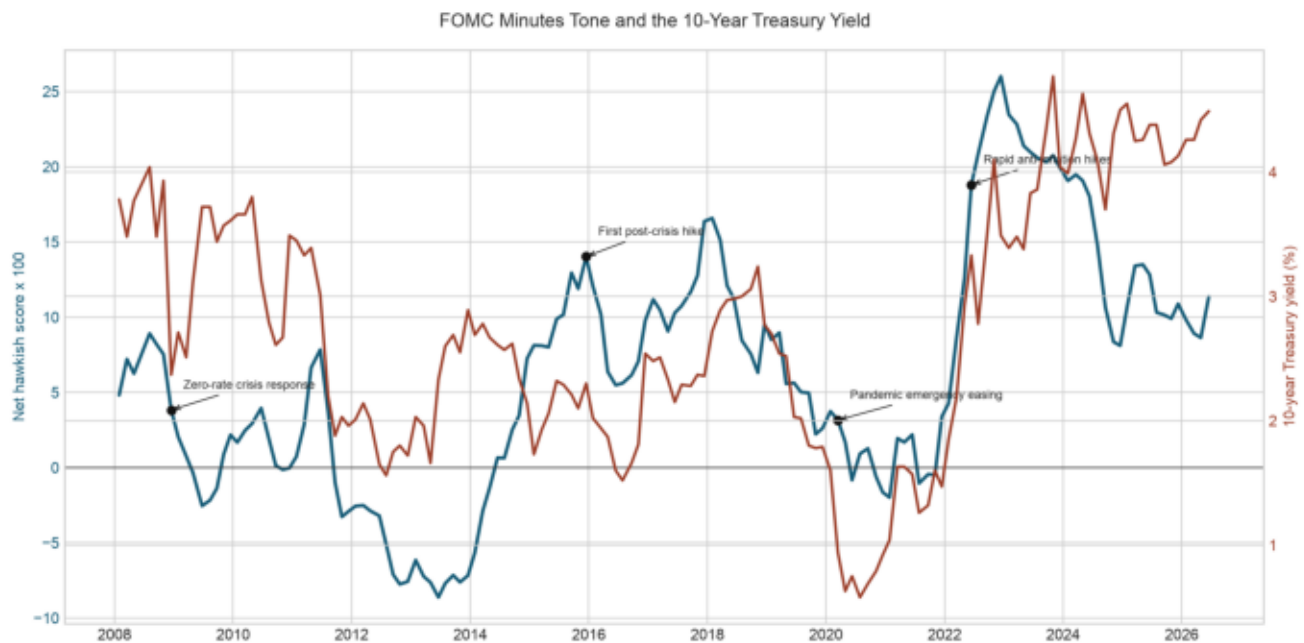
DATA SOURCES AND PREPROCESSING

- 147 FOMC minutes from 2008 through June 2026, downloaded from the Board of Governors' public meeting pages.
- FRED DGS10 and DGS2 Treasury series, aligned to the latest observation on or before each meeting date; the corresponding minutes are released later.
- HTML cleaned to readable text and split into sentences.

CLASSIFICATION AND AGGREGATION

- Count hawkish terms such as inflation pressure, tightening, restrictive, and rate increase; count dovish terms such as accommodation, easing, downside risk, slack, and asset purchases.
- Assign the direction with the larger count; classify ties as neutral.
- Compute (hawkish sentences – dovish sentences) / total sentences, then apply a three-meeting moving average.
- Fine-tune FinBERT as a three-class comparison using dictionary-generated weak labels and evaluate both classifiers against the same independent human audit.

4. Results — Main Time Series



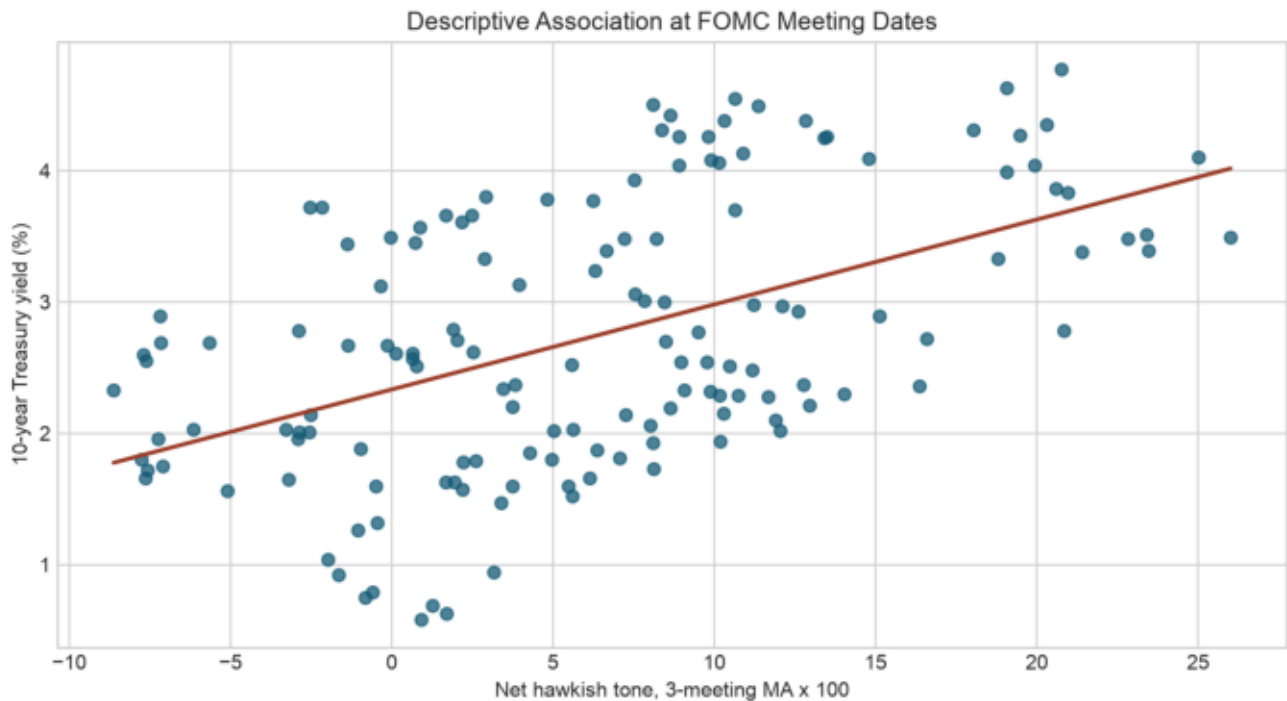
WHAT THE FIGURE SHOWS

The smoothed hawkish-dovish score rises during tightening cycles and shifts around major policy regimes.

MARKED POLICY REGIMES

- 2008 zero-rate crisis response
- 2015 first post-crisis rate hike
- 2020 pandemic emergency easing
- 2022 rapid anti-inflation hiking cycle

4. Results — Exploratory Market Association



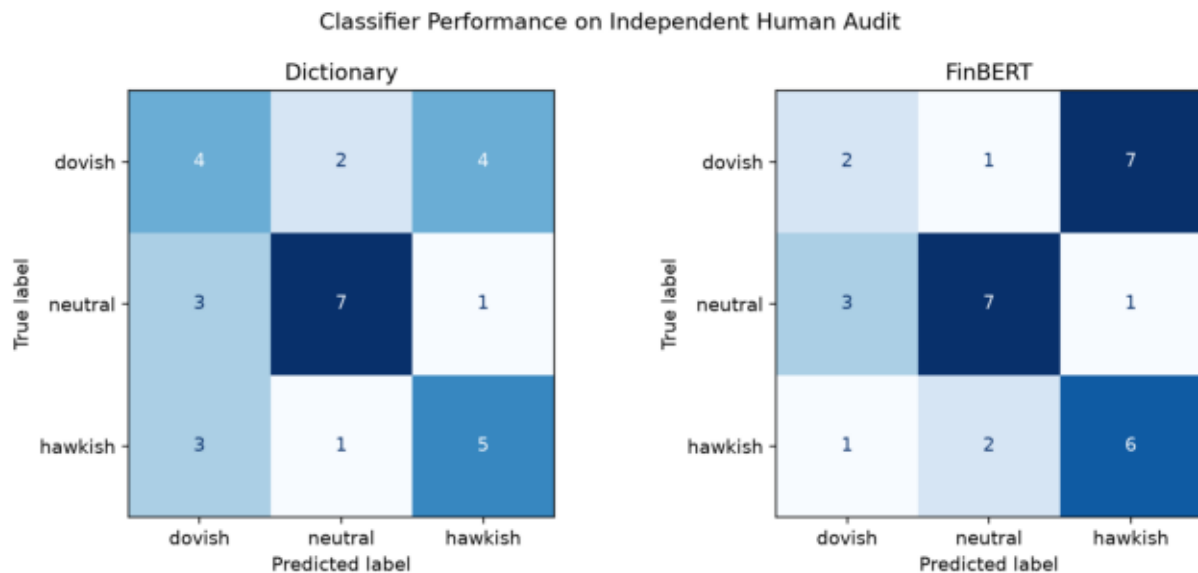
INTERPRETATION

The scatterplot shows a positive descriptive association between the three-meeting-smoothed tone index and the 10-year Treasury yield at FOMC meeting dates.

IMPORTANT QUALIFICATION

Both measures may vary with broader monetary-policy regimes. The pattern does not establish that the minutes caused changes in yields.

4. Results — Independent Classifier Evaluation



AUDIT RESULT

On the independent 30-sentence human audit, the dictionary baseline achieved 53.3% accuracy and 0.531 macro-F1, while the weakly supervised FinBERT model achieved 50.0% accuracy and 0.479 macro-F1. The observed accuracy difference was one correctly classified sentence, and the audit is too small to establish a meaningful performance difference.

TRACEABLE AUDIT EXAMPLES

- Dovish: "The staff still expected that the pace of economic activity through 2011 would be sufficient..." (human dovish; dictionary dovish; FinBERT dovish)
- Neutral: "A Committee decision to implement a program of agency MBS sales would be announced well in..." (human neutral; dictionary neutral; FinBERT neutral)
- Hawkish: "Inflation firmed in recent months, and most participants anticipated that it would continue..." (human hawkish; dictionary hawkish; FinBERT hawkish)
- Model Disagreement: "In addition, some participants noted signs that the finances of some households - especially..." (human hawkish; dictionary dovish; FinBERT neutral)

5. Discussion

CLASSIFIER EVALUATION

On the independent 30-sentence human audit, the dictionary baseline achieved 53.3% accuracy and 0.531 macro-F1, while the weakly supervised FinBERT model achieved 50.0% accuracy and 0.479 macro-F1. The observed accuracy difference was one correctly classified sentence, and the audit is too small to establish a meaningful performance difference. This illustrates a central limitation of weak supervision: a model trained primarily from dictionary-generated labels may reproduce the teacher's assumptions without gaining greater agreement with independent human judgments.

MARKET INTERPRETATION

The sample contains 147 FOMC minutes from 2008-01-30 to 2026-06-17. The contemporaneous descriptive correlation between the three-meeting-smoothed tone index and DGS10 is $r=0.510$. Across the eight combinations formed from two Treasury-yield series and four tone lags, the largest observed correlation was for DGS2 with tone lagged by 2 meetings ($r=0.806$). This was an exploratory comparison rather than a prespecified or uniquely validated model specification. The larger DGS2 association is consistent with the 2-year yield's sensitivity to near-term monetary-policy conditions, but it may also reflect persistent common trends and broad policy regimes. Market correlation does not validate sentence-label accuracy.

LIMITATIONS

- Dictionary models can miss negation, conditional language, and context.
- The human audit is small, stratified by dictionary prediction, and labeled by one annotator.
- Three-meeting smoothing creates overlapping observations; tone and yields are serially persistent.
- Pearson p-values do not adjust for dependence or selection across eight comparisons.
- Meeting-date yields precede the later minutes releases, so this is neither causal analysis nor a minutes-release event study.

POTENTIAL IMPROVEMENTS

- Expand the independently reviewed label set.
- Evaluate statements separately from minutes.
- Study release-date windows and futures surprises.
- Use time-series methods that account for persistence and multiple comparisons.

6. Conclusion

ACHIEVEMENTS

- Built a reproducible workflow for public-data acquisition, sentence-level policy-tone classification, document aggregation, independent human-audit evaluation, and exploratory market visualization.
- Measured 53.3% dictionary accuracy and 50.0% weakly supervised FinBERT accuracy on the 30-sentence audit; the sample is too small to establish a meaningful difference.

LESSONS LEARNED

- Independent labels are necessary; agreement with dictionary-generated training labels is not independent accuracy.
- A pretrained model does not necessarily outperform a transparent baseline when its supervision inherits the baseline's assumptions.
- Market association is a separate descriptive analysis, not classifier ground truth.

INDIVIDUAL CONTRIBUTIONS — JINYU CHEN

- Built the original data-collection and preprocessing pipeline, dictionary classifier, document-level tone and Treasury-yield analysis, and FinBERT fine-tuning workflow.
- Created the original figures, notebook, report, and presentation materials.

INDIVIDUAL CONTRIBUTIONS — DYLAN ULLRICH

- Added the reproducible development environment and clarified the saved FinBERT artifacts' weak-label provenance.
- Created and manually labeled the 30-sentence audit; implemented classifier evaluation and audit-provenance checks.
- Corrected the market-analysis interpretation; filmed the video; and updated the README, notebook, report, slides, script, figures, data, and final submission package.

7. References

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